

Execution Times

scenario	seconds
ORIGINAL	3.24675
OPTIMIZED	1.05495

Block I/O Summary (validation steps excluded)

scenario	block_input_operations	block_output_operations	total_block_io
OPTIMIZED	0	0	0
ORIGINAL	0	29,952	29,952

Final Benchmark Results (validation steps excluded)

original_seconds	optimized_seconds	runtime_seconds_saved	runtime_savings_pct	original_block_input	original_block_output
3.247	1.055	2.192	67.51	0	29,952

original_total_io	optimized_block_input	optimized_block_output	optimized_total_io	block_io_saved	io_savings_pct
29,952	0	0	0	29,952	100.00

Data sample from Countries POP GDP (Not Optimized and Optimized)

COUNTRY_CODE	SHORT_NAME	SORTABLE_NAME	LONG_NAME	ALPHA_2_CODE	REGION	INCOME_GROUP	WB_2_CODE
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
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COUNTRY_NAME	YEAR	POPULATION	POP_GROWTH	GDP	GDP_PCT_CHANGE	LAGGED_GDP	WORLD_GDP
Aruba	2000	89101	0.025392	1873452514.0	0.08738	1722905027.9	2.5005E14
Aruba	2002	91781	0.011947	1961843575.4	0.03448	1896456983.2	2.5936E14
Aruba	2004	93540	0.009010	2254830726.3	0.10309	2044111731.8	3.326E14
Aruba	2006	95606	0.011816	2469782681.6	0.04651	2360017318.4	4.0015E14
Aruba	2008	97996	0.012414	2843024581.0	0.06176	2677641340.8	5.1234E14
Aruba	2010	100341	0.011315	2453597206.7	-0.03923	2553793296.1	5.4003E14
Aruba	2012	102112	0.008102	2615208379.9	-0.00859	2637859217.9	6.2086E14
Aruba	2014	103594	0.006916	2790849720.7	0.02310	2727849720.7	6.6136E14
Aruba	2016	104874	0.005901	2983635195.5	0.00700	2962907262.6	6.2894E14
Aruba	2018	105962	0.004948	3276184357.5	0.05942	3092429050.3	7.1622E14

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ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
ABW	Aruba	Aruba	Aruba	AW	Latin America & Caribbean	High income	AW
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Original_Pop_GDP_Table
3192

New_Pop_GDP_Table
3192

Data sample from Country Lookup (Not Optimized and Optimized)

ISOALPHA3	COUNTRY_CODE	IDNAME	COUNTRY_NAME
ABW	ABW	Aruba	Aruba
AFG	AFG	Afghanistan	Afghanistan
AGO	AGO	Angola	Angola
ALB	ALB	Albania	Albania
AND	AND	Andorra	Andorra
ARE	ARE	United Arab Emirates	United Arab Emirates
ARG	ARG	Argentina	Argentina
ARM	ARM	Armenia	Armenia
ASM	ASM	American Samoa	American Samoa
ATG	ATG	Antigua and Barbuda	Antigua and Barbuda

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AFG	AFG	Afghanistan	Afghanistan
AGO	AGO	Angola	Angola
ALB	ALB	Albania	Albania
AND	AND	Andorra	Andorra
ARE	ARE	United Arab Emirates	United Arab Emirates
ARG	ARG	Argentina	Argentina
ARM	ARM	Armenia	Armenia
ASM	ASM	American Samoa	American Samoa
ATG	ATG	Antigua and Barbuda	Antigua and Barbuda

Original_Country_Lookup
214

New_Country_Lookup
214